KEPLER'S THIRD LAW ON LOG–LOG AXES

The planets draw a power law

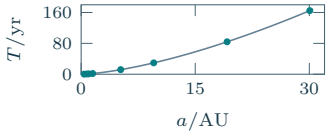
Observations reveal the slope. Dimensions explain the exponent. Dynamics supplies the coefficient.

1 Planetary data

Planet	a/AU	T/yr
Mercury	0.387	0.241
Venus	0.723	0.615
Earth	1.00	1.00
Mars	1.52	1.88
Jupiter	5.20	11.86
Saturn	9.54	29.46
Uranus	19.19	84.01
Neptune	30.06	164.82

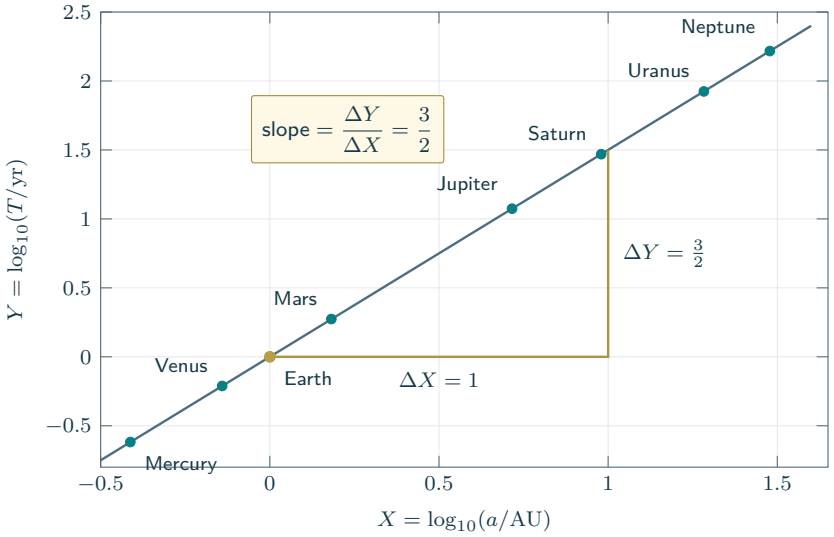
Rounded orbital values.

Linear axes



2 The same data, straightened

One decade across corresponds to one and a half decades up.



Data fit: slope $\simeq 1.500$. Reference line: $Y = \frac{3}{2}X$.

3 A power law becomes a straight line

$$\frac{T}{\mathrm{yr}} = K\left(\frac{a}{\mathrm{AU}}\right)^{p} \quad \Longleftrightarrow \quad \boxed{Y = pX + \log_{10} K}.$$

For planets around Sun, $p \simeq 3/2$ and $K \simeq 1$. Thus $T \propto a^{3/2}$, or equivalently $\boxed{T^2 \propto a^3}$. Here a is the ellipse's **semimajor axis**.

4 Dimensions predict the exponent

Hold the orbit's shape fixed, neglect the planet's mass, and suppose only a and GM set the period. Here M is the central mass.

$$T = C\,a^{\alpha}(GM)^{\beta}, \qquad [GM] = \mathsf{L}^3\mathsf{T}^{-2}, \qquad [T] = \mathsf{L}^{\alpha+3\beta}\mathsf{T}^{-2\beta}.$$

Matching length and time powers gives

$$\alpha + 3\beta = 0, \qquad -2\beta = 1 \quad \Longrightarrow \quad \boxed{T = C\sqrt{\frac{a^3}{GM}}}.$$

The $3/2$ exponent follows from these model inputs. For ellipses, dimensions could still allow C to depend on the dimensionless eccentricity e; the coefficient needs dynamics.

5 Newton supplies the coefficient

For a circular test-particle orbit, gravitational force provides the centripetal acceleration:

$$\frac{GMm}{a^2} = m\frac{4\pi^2 a}{T^2} \quad \Longrightarrow \quad \boxed{T = 2\pi\sqrt{\frac{a^3}{GM}}}.$$

The full elliptical solution gives the same result with a as the semimajor axis: $C(e) = 2\pi$, independent of eccentricity.

6 The architecture of scale

$$\boxed{a \mapsto sa \quad \Longrightarrow \quad T \mapsto s^{3/2}T} \qquad \text{at fixed central mass.}$$

$\times 2 \;\rightarrow\; \times 2.83 \qquad \times 4 \;\rightarrow\; \times 8 \qquad \times 10 \;\rightarrow\; \times 31.6$

Units and central mass shift the intercept. The exponent records the response to a change of scale.

Rounded data: NASA planetary fact sheets and OpenStax, *Astronomy 2e*, Table 3.2. Orbital law: NASA, *Orbits and Kepler's Laws*. Graphs and dimensional argument constructed from the stated data and assumptions.